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Differentially Private CutMix for Split Learning with Vision Transformer

Seungeun Oh[1], Sihun Baek[1], Hyelin Nam[1], Seong-Lyun Kim[1], Jihong Park[2], Praneeth Vepakomma[3], Ramesh Raskar[3], Mehdi Bennis[4]
[1] Yonsei University [2] Deakin University [3] Massachusetts Institute of Technology [4] University of Oulu

연세대학교 MIT

Paper

Problem

In split learning with a Vision Transformer, the cut-layer **smashed data** barely differs from the raw input — no pooling, no convolution — **leaking privacy** and inflating cost.

> *No pooling, no convolution: a ViT's smashed data barely distorts the input, so it leaks like the raw image.*

Motivation

- ViT is displacing CNN in vision, so privacy-preserving distributed training must be rethought for pooling-free, patch-level transformer architectures.
- Its cut-layer output barely distorts the input, so regularizing smashed data works as well as regularizing the raw image — but leaks privacy.
- Three ViT traits — low distortion, noise-robust attention, and patch-level operation — point to a single **patch-scale regularizer** across clients.

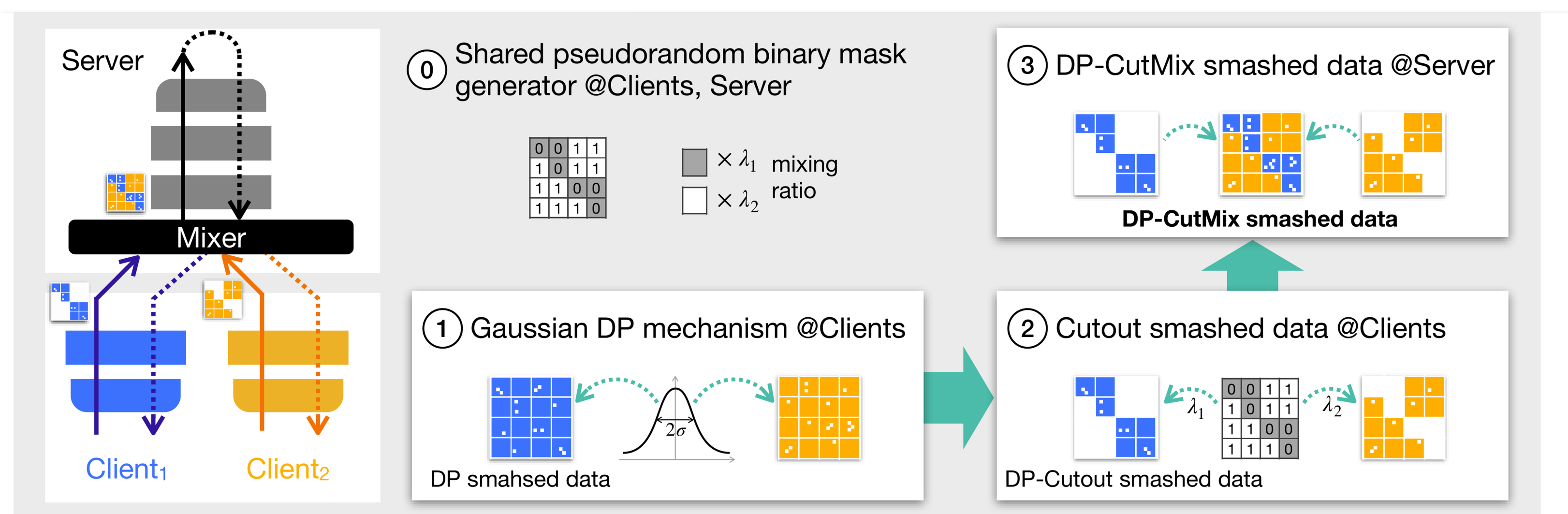
Global self-attention makes ViT robust to noise on part of the image — ideal for Cutout / CutMix-style regularization, so masking whole patches costs little accuracy.

Method

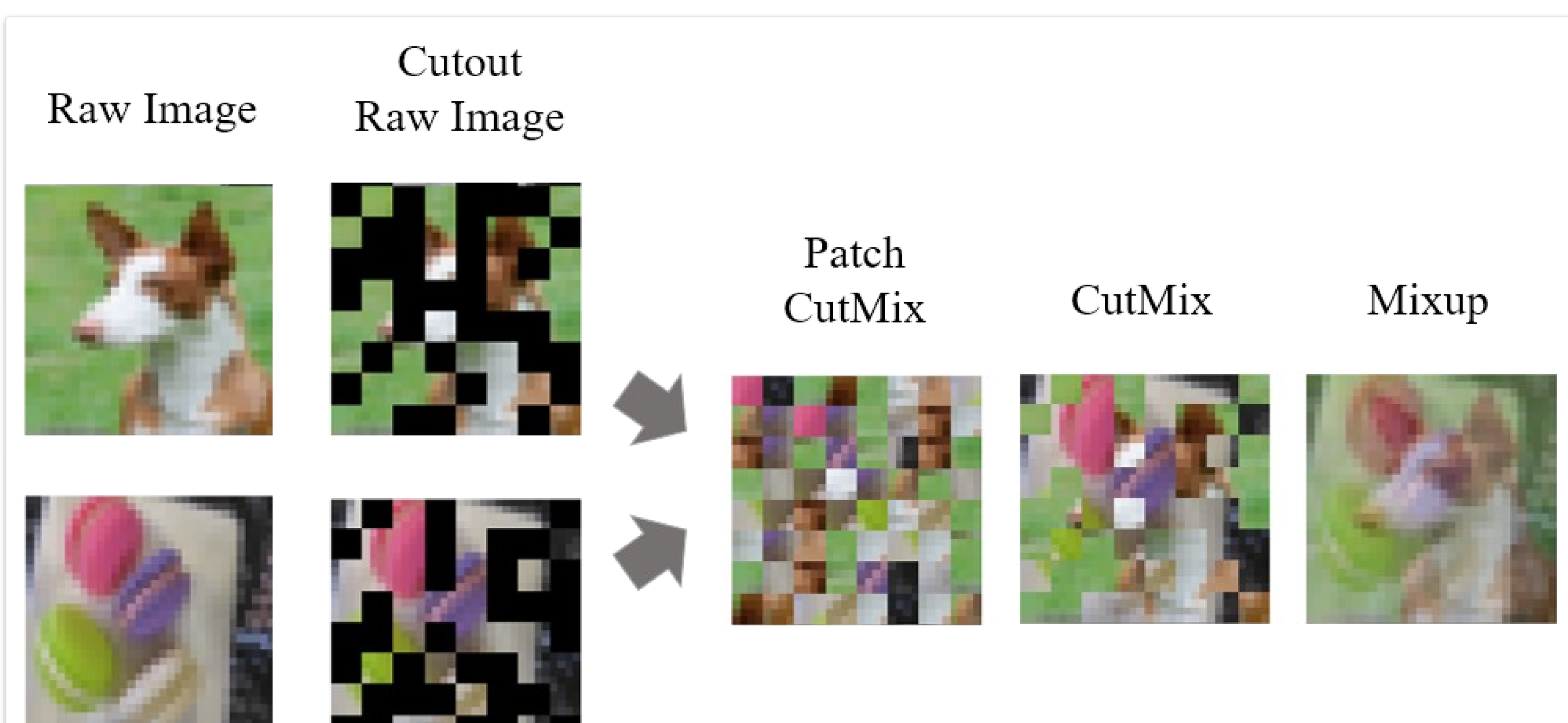
- A **mixer** shares a pseudorandom binary mask M_i (Dirichlet ratios λ_i, $\Sigma\lambda_i = 1$) to each client.
- Clients mask smashed data (**Cutout**) and add Gaussian noise; the server sums complementary patches → **DP-CutMix**, then propagates as in vanilla SL.

$$\tilde{s}'_{i,j} = \bar{s}'_i + \bar{s}'_j, \qquad \varepsilon_{\mathrm{Mix}}(\alpha) \le \varepsilon_{\mathrm{CutMix}}(\alpha) \le \varepsilon_o(\alpha)$$

Randomized inter-client CutMix provably tightens the Rényi-DP budget.



DP-CutMixSL: a shared mask → Gaussian-DP Cutout at clients → the server stitches surviving patches into DP-CutMix smashed data.



Interpolation schemes on raw images vs. smashed data: patch-level Cutout, Patch CutMix, CutMix, Mixup.

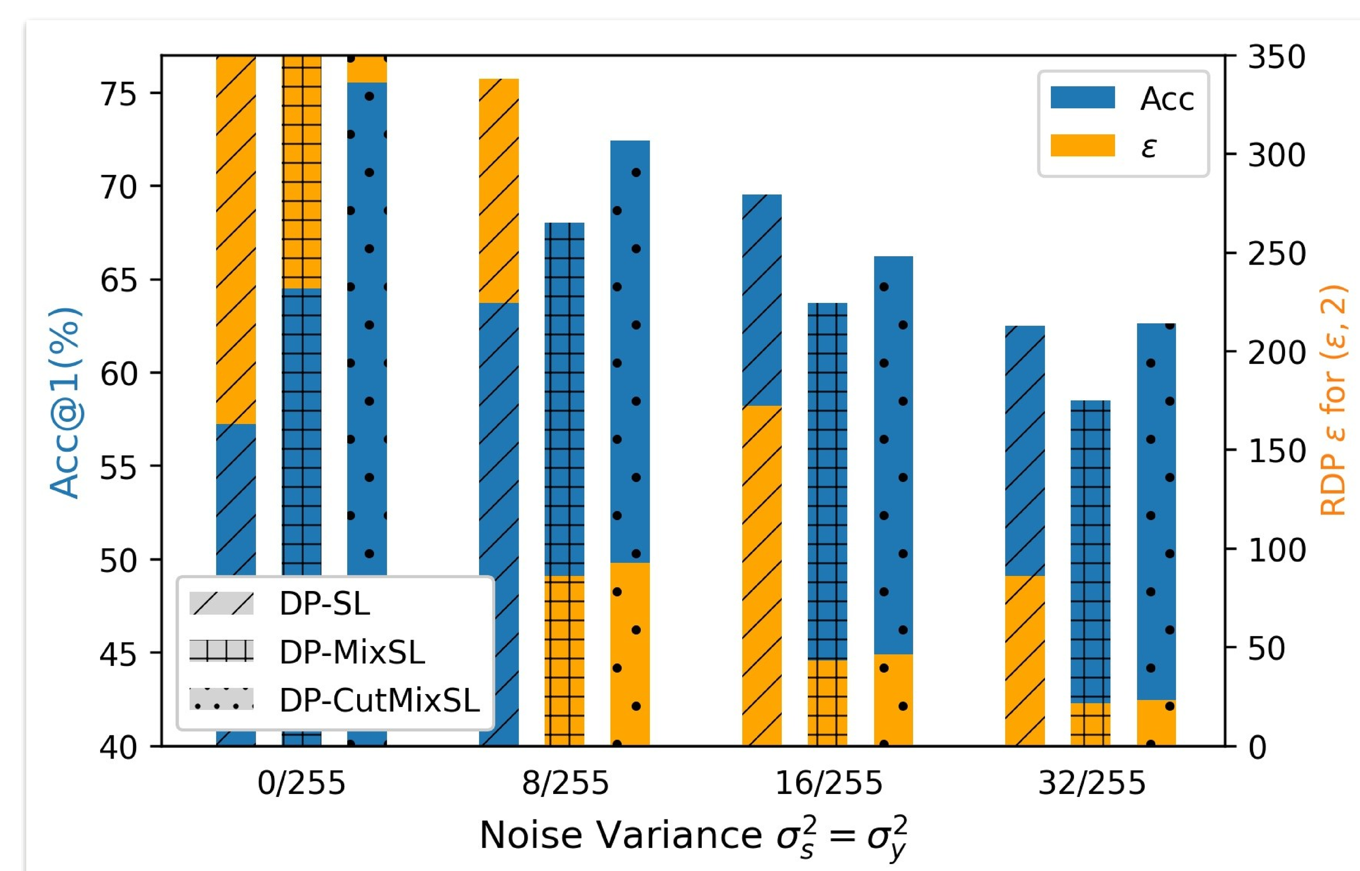
Dataset / Benchmark

CIFAR-10 · Fashion-MNIST · ViT-Tiny · PiT-Tiny · VGG-16 · |C| = 10 clients · cut after embedding

PiT-Tiny inserts a pooling layer, bridging pure ViT and CNN; the cut-layer sits right after patch embedding. Standalone, SL and SplitFed are baselines. RDP computed at patch size N = 64, sensitivity Δ = 0.2, uniform $\lambda_i = 1/n$, order α = 2.

Key Results

CIFAR-10 TOP-1 (|C|=10)	VIT-TINY	PIT-TINY	VGG-16
SL	57.21	52.28	62.62
SplitFed	67.88	55.63	63.98
CutMixSL (ours)	**73.77**	**71.26**	**67.53**



Accuracy & RDP ε vs. noise variance (CIFAR-10): DP-CutMixSL leads accuracy at nearly every level, at a tighter privacy budget than DP-SL.

Headline Numbers

73.77% Top-1 · CIFAR-10 · ViT-Tiny
+16.56 pp over plain SL

71.26% PiT-Tiny top-1

89.75% F-MNIST · ViT

~8× harder to invert

Ablation Study

- **DP-CutMixSL** wins accuracy at almost every noise level.

ROBUST

Recon. MSE: Cutout > Patch CutMix > Mixup — 8× harder to invert.

Takeaway

For Vision Transformers, DP-CutMixSL breaks the privacy–accuracy trade-off: mixing noisy masked patches across clients both tightens the DP guarantee and improves accuracy.

Patch-level, global-attention ViT makes patch CutMix work where it would hurt a CNN — a privacy regularizer for the transformer era, with the privacy gain provably bounded by a Mixup counterpart.